LEARN LENS

From Mistakes to Mastery

AI-Powered Adaptive Learning & Learning Verification System

Concept-Level Assessment

Verify Understanding

Personalized Education

Prasunethon 2.0 — Round 2 Project Submission

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PROBLEM

Getting an Answer Right ≠ Understanding the Concept

Students can complete questions and still have underlying weaknesses.

Current learning systems often follow:

Learn → Quiz → Correct/ Wrong → Score

Core Problem

Current systems measure answers, but miss continuous understanding.

Missing

- Cause of the mistake
- Repeated mistakes
- Concept understanding
- Independent problem-solving

INSIGHT

Learning Should Be Verified, Not Assumed

A learner shouldn't move forward simply because they completed a lesson.

LearnLens evaluates understanding through:

Recall → Explain → Predict → Implement → Debug → Apply

Existing Approach

Lesson → Quiz → Score → Next

LearnLens Approach

Learn → Attempt → Collect Evidence → Analyze → Verify → Adapt

Key Insight

A correct answer is only one signal of understanding.

Understanding, backed by evidence.

THE SOLUTION

LearnLens — From Learning to Proof of Understanding

LearnLens builds a continuously evolving understanding profile for each learner.

1 Step 1 — Concept

2 Step 2 — Theory + Coding Challenge

3 Step 3 — Student Response

4 Step 4 — Mistake & Evidence Analysis

5 Step 5 — Knowledge Fingerprint

6 Step 6 — Targeted Challenge

7 Step 7 — Proof-of-Learning

7 Step 7 — Advance or Reinforce

Learning paths adapt to what students actually understand.

HOW IT WORKS

The LearnLens Learning Loop

Every attempt becomes learning evidence

- LEARN CONCEPT
- ↓
- THEORY + CODING ATTEMPT
- ↓
- CODE EXECUTION / RESPONSE ANALYSIS
- ↓
- MISTAKE DETECTION
- ↓
- KNOWLEDGE FINGERPRINT
- ↓
- TARGETED CHALLENGE
- ↓
- PROOF-OF-LEARN
- ↓
- ADAPT NEXT CONCEPT

CORE FEATURES

From Learning Activity to Learning Intelligence

01 Adaptive Learning Roadmap

Basic → Intermediate → Advanced, with progression based on demonstrated understanding.

02 Code Execution & Mistake Tracking

Record execution errors, wrong outputs, attempts and recurring patterns.

03 Proof-of-Learn

Require evidence of understanding before considering a concept mastered.

04 Theory + Coding Assessment

Explain, predict, implement, modify and debug concepts.

05 Knowledge Fingerprint

Profile understanding across Recall, Explain, Predict, Implement, Debug and Apply.

06 Adaptive Challenges

Give targeted challenges based on the learner's specific weakness.

ADAPTIVE LEARNING ROADMAP

The Roadmap Adapts to the Learner

Example

Variables →Conditions → Loops →Functions → Arrays → OOP

But progression is not based only on completion.

Example

Student A

Loops → Strong

Functions → Weak

↓

Receives additional Functions practice

↓

Reassessment

↓

Moves forward

The roadmap changes according to demonstrated understanding.

THEORY + CODING + MISTAKE INTELLIGENCE

ONE CONCEPT. MULTIPLE FORMS OF EVIDENCE.

For every concept, LearnLens can assess:

Theory

Explain → Predict → Why?

Coding

Write → Modify → Debug → Solve

Mistake Intelligence

The system records:
Error → Type → Concept → Recurrence → Improvement

Example

Loop boundary error detected 3 times
↓
Likely weakness: boundary-condition reasoning
↓
Targeted practice

KNOWLEDGE FINGERPRINT

1 Measure How the Learner Understands

Instead of:

Recursion = 70%

2 LearnLens builds a Knowledge Fingerprint:

SKILL PROFICIENCY

Recall	🟢	■■■■■■■■■	Strong
Explain	🟢	■■■■■■■■	Strong
Predict	🟡	■■■■■■	Moderate
Implement	🟡	■■■■■	Moderate
Debug	🔴	■■■	Weak
Apply	🔴	■■	Weak

Progress

🟢 Strong 🟡 Moderate 🔴 Weak

3 Result

The learner understands recursion theory, but struggles to debug and apply it to unfamiliar problems.

PROOF-OF-LEARN

DON'T ASSUME MASTERY. VERIFY IT.

After learning a concept, LearnLens creates a new challenge.

📖 LEARNED EXAMPLE
↓
💡 EXPLAIN
↓
🔮 PREDICT
↓
🛠️ IMPLEMENT
↓
🐞 DEBUG
↓
🚀 APPLY TO A NEW PROBLEM
↓
🏆 PROOF-OF-LEARN

If understanding is sufficient:

Concept Mastered → Next Concept

If understanding is insufficient:

Weakness → Targeted Reinforcement

Learning is considered demonstrated only when the learner can apply the concept beyond the original example.

ADAPTIVE CHALLENGES

SAME TOPIC. DIFFERENT LEARNING PATH.

STUDENT A

FUNCTIONS
↓
THEORY → WEAK
↓
🎯 CHALLENGE
Explain parameters
and return values
↓
🔄 REASSESSMENT

STUDENT B

FUNCTIONS
↓
THEORY → STRONG
↓
CODING → WEAK
↓
🎯 CHALLENGE
Write & debug
function-based problems
↓
🔄 REASSESSMENT

Different weakness ➡ Different challenge

CORE INNOVATION

From "What Did You Score?" to "What Can You Actually Do?"

Traditional learning	Recursion → 70%
LearnLens	RECURSION

Dimension		Understanding
• Recall	→	Strong
• Explain	→	Strong
• Predict	→	Moderate
• Implement	→	Moderate
• Debug	→	Weak
• Apply	→	Weak

AI-generated Insight - Result

"You understand recursion theory, but struggle to debug and apply it to unfamiliar problems."

Innovation

Knowledge Fingerprint + Proof-of-Learn = Learning Verification

A concept is not considered mastered until the learner demonstrates understanding.

ADAPTIVE LEARNING

Same Topic. Different Learning Path.

Student A

Theory Weak

Loops

Concept Understanding → Weak

Coding → Strong

↓

Targeted Challenge

↓

Reassessment

Student B

Coding Weak

Loops

Concept Understanding → Strong

Coding → Weak

↓

Targeted Challenge

↓

Reassessment

LearnLens adapts the next challenge to the learner's actual weakness.

USER JOURNEY

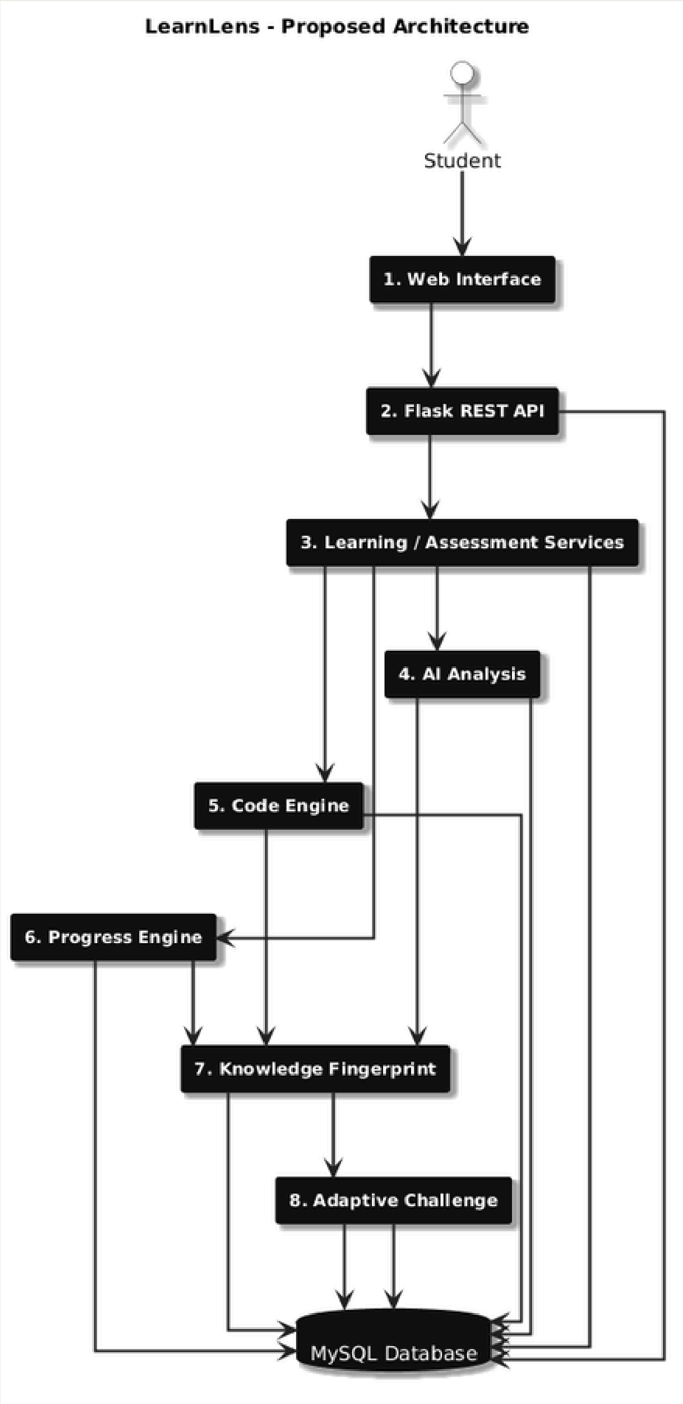
One Mistake Can Change the Learning Path

Learn · Explain · Implement · Execute · Analyze · Target · Prove · Progress

Every attempt makes the learner profile more informative.

TECHNOLOGY & ARCHITECTURE

Technically Feasible MVP Architecture



AI handles analysis and personalization.
Deterministic code handles scoring, storage, validation and progression rules.

Stack

Frontend

HTML • CSS • JavaScript

Backend

Python • Flask • REST API

Database

MySQL

AI

LLM API • Structured Response Validation

Coding Evaluation

Sandboxed Code Execution

Authentication

Password Hashing • Sessions

IMPACT & UNIQUENESS

From AI-Assisted Learning to AI-Verified Learning

Traditional LMS

- Completion
- Scores
- Progress

Generic AI Tutor

- Answers
- Explanations
- Examples

LearnLens

- Understanding Evidence
- Mistake Patterns
- Knowledge Fingerprint
- Adaptive Challenges
- Proof-of-Learn

Students

- Precise weaknesses
- Targeted practice
- Stronger independent problem-solving

Impact

Educators

- Concept-level insights
- Recurring misconceptions

Difference

Don't just measure the answer. Verify the understanding.

ROADMAP

20-23 AUGUST | BUILD → VALIDATE → DEMONSTRATE

20 AUG — FOUNDATION

- Roadmap
- Authentication
- Concepts
- Questions
- Attempts

21 AUG — EVIDENCE

- Code Execution
- Mistake Tracking
- AI Analysis

22 AUG — INNOVATION

- Knowledge Fingerprint
- Adaptive Challenges
- Proof-of-Learn

23 AUG — VALIDATION & DELIVERY

- Reassessment
- Progress
- Testing
- Deployment
- Demo

ROUND 2 MVP

Learn
Attempt
Analyze
Fingerprint
Target
Verify
Adapt

Future Scope

PDF/RAG • PYQ Analysis • Teacher Analytics • Exam Priority

LearnLens

Every mistake becomes a learning signal.

Thank You

Team Nova - Maria Dorthy

Prasunethon 2.0 — Round 1 Idea Submission